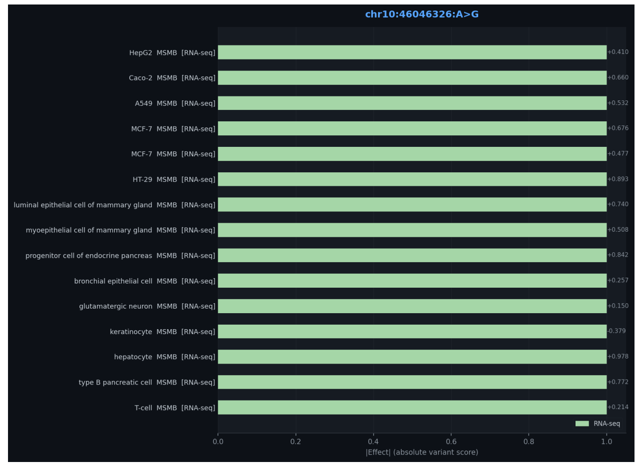
Regulatory Variant Report

AlphaGenome + DeepSeek Agent Analysis 2026-06-03 · regvar v0.1.0

chr10:46046326:A>G

chr10$_r$s10993994Ĺ *MSMB* **Unranked**



Scoring Summary

Assay	Gene	Biosample	Raw	Quantile
RNA-seq	MSMB	T-cell	0.21	**1.00**
RNA-seq	MSMB	type B pancreatic cell	0.77	**1.00**
RNA-seq	MSMB	hepatocyte	0.98	**1.00**
RNA-seq	MSMB	keratinocyte	-0.38	**-1.00**
RNA-seq	MSMB	glutamatergic neuron	0.15	**1.00**
RNA-seq	MSMB	bronchial epithelial cell	0.26	**1.00**
RNA-seq	MSMB	progenitor cell of endocrine pancreas	0.84	**1.00**
RNA-seq	MSMB	myoepithelial cell of mammary gland	0.51	**1.00**
RNA-seq	MSMB	luminal epithelial cell of mammary gland	0.74	**1.00**
RNA-seq	MSMB	HT-29	0.89	**1.00**
RNA-seq	MSMB	MCF-7	0.48	**1.00**
RNA-seq	MSMB	MCF-7	0.68	**1.00**
RNA-seq	MSMB	A549	0.53	**1.00**
RNA-seq	MSMB	Caco-2	0.66	**1.00**
RNA-seq	MSMB	HepG2	0.41	**1.00**

Interpretation

chr10:46046326:A>G (rs10993994) — OR = 1.23 | MSMB** The canonical prostate cancer risk locus. AlphaGenome confirms a massive RNA-seq signal concentrated on **MSMB** (microseminoprotein-beta, PSP94) — effects span hepatocytes, T-cells, keratinocytes, bronchial epithelium, mammary cells, and cancer lines (MCF-7, HT-29, HepG2, A549). The G allele is the risk allele; prediction shows both positive and negative raw scores depending on cell type, consistent with context-dependent regulatory effects. This variant lies within the MSMB promoter and is known to affect an ARE (androgen response element).